

Smartwatch for Real-Time Health and Location Monitoring

Tasrina Hussan, Nadia Zaman, Jannatul Maowa, Nafisa Nawar
Department of Electronics and Telecommunication Engineering
Chittagong University of Engineering & Technology
Chattogram, Bangladesh

Abstract—This paper presents a smartwatch system for real-time health monitoring, location tracking, and interactive display faces. The system integrates ESP32 as the main controller, with sensors such as MAX30102 for heart rate and SpO₂, MPU6050 for motion detection, GPS for location, and RTC for accurate timekeeping. The smartwatch features multiple interactive faces including digital, analog, health, location-weather, and menu faces with sub-functions like alarm, stopwatch, and settings. Data collected is transmitted to a mobile or web interface for remote monitoring, enabling continuous observation of elderly or health-conscious individuals. The proposed system is low-cost, reliable, and user-friendly, providing an effective wearable solution for personal healthcare management.

Index Terms—Smartwatch, ESP32, MAX30102, MPU6050, Health Monitoring, GPS, OLED, Wearable Device

I. INTRODUCTION

The rapid advancement of the wearable technology has transformed healthcare by enabling personalized, real-time monitoring. Traditionally, health tracking required clinical visits with bulky equipment and professional supervision. Modern wearable devices, however, integrate high-performance microcontrollers with compact biometric sensors, allowing continuous, non-invasive monitoring of vital signs during daily activities. This capability is particularly important for elderly and disabled individuals, who need consistent observation to prevent emergencies such as cardiac events or accidental falls.

This project presents the design and implementation of a multifunctional smartwatch based on the ESP32 platform. Unlike conventional smartwatches, which are often limited in medical functionality, this system combines lifestyle features with clinical-grade data acquisition. Using the MAX30102 pulse oximeter and heart-rate sensor, the device can detect subtle variations in blood oxygen saturation (SpO₂) and heart rate, key indicators for conditions like obstructive sleep apnea and cardiovascular irregularities. The MPU6050 inertial measurement unit (IMU) further enables step counting and fall detection, providing additional safety and activity monitoring.

The smartwatch also features a sophisticated user interface on a 1.3-inch OLED display, organized into multiple “faces” such as analog time, GPS-based location tracking, weather updates, and utility functions including alarms and stopwatches. A GPS module and Real-Time Clock (RTC) ensure accurate temporal and spatial information, allowing reliable location mapping and precise timekeeping for emergency alerts and daily scheduling.

Overall, this implementation demonstrates a cost-effective, scalable, and highly functional wearable solution that integrates health monitoring, safety features, and everyday utility, providing continuous care without compromising the convenience and versatility of a modern smartwatch.

II. SYSTEM FRAMEWORK

The proposed smartwatch system integrates health monitoring, activity tracking, and IoT connectivity into a compact wearable platform. The ESP32 microcontroller serves as the central unit, managing sensor data acquisition, signal processing, display control, and wireless communication. Key sensors include the MAX30102 pulse oximeter for heart rate and blood oxygen saturation (SpO₂), and the MPU6050 accelerometer and gyroscope for step counting and fall detection. A GPS module and Real-Time Clock (RTC) provide accurate location and timekeeping for navigation, alarms, and scheduling.

A 1.3-inch OLED display presents multiple watch faces, including digital and analog time, health metrics, location and weather information, and utility functions such as alarms and stopwatches. Users interact with the device through tactile buttons for seamless navigation. Sensor data is processed in real time, with critical events like abnormal SpO₂ levels or falls immediately communicated via Wi-Fi to a cloud server or mobile application for remote monitoring.

The system is designed to be modular, energy-efficient, and scalable. Data flows from sensors to the ESP32 for processing and display, while simultaneously being transmitted for remote access. This framework ensures reliable real-time health tracking, emergency alerting, and user-friendly operation, making it particularly suitable for elderly and disabled individuals.

III. CONSTITUENT SUBMODULES

The proposed smartwatch system consists of several integrated hardware submodules, each performing a critical function in monitoring, processing, and displaying data. The main submodules include the ESP32 microcontroller, MAX30102 pulse oximeter, MPU6050 accelerometer and gyroscope, GPS module, Real-Time Clock (RTC), OLED display, and user input buttons.

A. ESP32 Microcontroller

The ESP32 is a low-cost, high-performance microcontroller with a dual-core processor, integrated Wi-Fi, and Bluetooth

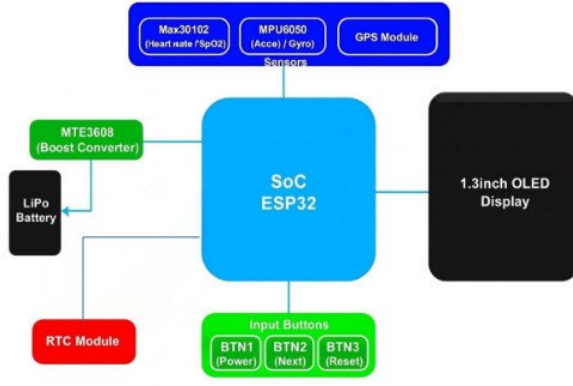


Fig. 1. Smartwatch Hardware Architecture

capabilities. It serves as the main processing unit of the smartwatch, handling data acquisition, signal processing, display control, and IoT communication. The ESP32 supports multiple I/O interfaces, including I2C, SPI, UART, and PWM, allowing seamless integration with sensors and display modules. Its low-power sleep modes and flexible clock configuration make it ideal for wearable applications.

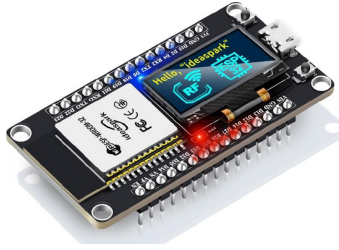


Fig. 2. ESP32 Microcontroller Board

B. MAX30102 Pulse Oximeter and Heart Rate Sensor

The MAX30102 is an integrated pulse oximeter and heart-rate sensor. It uses red and infrared LEDs along with a photodetector to measure blood oxygen saturation (SpO_2) and pulse rate. The sensor provides raw photoplethysmography (PPG) signals, which the ESP32 processes to calculate real-time heart rate and SpO_2 levels. It operates at 1.8V–3.3V and communicates via the I2C interface.

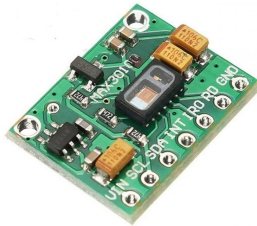


Fig. 3. MAX30102 Pulse Oximeter Sensor

C. MPU6050 Accelerometer and Gyroscope

The MPU6050 is a 6-axis MEMS sensor that combines a 3-axis accelerometer and a 3-axis gyroscope. It measures linear acceleration and angular velocity, enabling step counting, motion tracking, and fall detection. Data from the MPU6050 is transmitted to the ESP32 via I2C and processed to detect sudden changes in acceleration indicative of a fall.

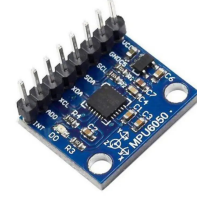


Fig. 4. MPU6050 Accelerometer and Gyroscope

D. GPS Module

The GPS module provides real-time geographic coordinates, speed, and altitude. It allows the smartwatch to track the user's location accurately for navigation and emergency alerts. The module communicates with the ESP32 through UART and is capable of updating position data every second.



Fig. 5. GPS Module

E. RTC Module (DS3231)

The Real-Time Clock module ensures precise timekeeping independent of the main processor. The DS3231 is highly accurate with a built-in temperature-compensated crystal oscillator and battery backup, maintaining correct time even when the device is powered off. It communicates with the ESP32 via I2C.

F. Boost Converter Module

The smartwatch is powered by a rechargeable Li-Po battery, which typically provides a nominal voltage of approximately 3.7V. However, several system components require a stable 5V supply for reliable operation. To meet this requirement, a DC-DC boost converter is used to step up the battery voltage to a regulated 5V output.

The boost converter efficiently increases the lower battery voltage to the required level while minimizing power loss,

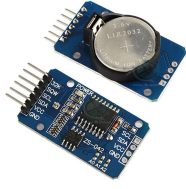


Fig. 6. RTC Module (DS3231)

making it suitable for wearable and battery-operated devices. The regulated 5V output is supplied to the ESP32 and peripheral modules, ensuring stable system performance even as the battery voltage varies during discharge. This power management approach improves reliability and allows the smartwatch to operate consistently over extended periods.



Fig. 7. DC-DC boost converter used to step up Li-Po battery voltage to 5V

G. OLED Display

The smartwatch features a 1.3-inch OLED display with I2C interface, providing high-resolution output for multiple watch faces, health metrics, location, and utility functions. The low-power display allows continuous monitoring without significant battery drain.



Fig. 8. 1.3-inch OLED Display

H. Push Buttons

Three tactile push buttons allow the user to interact with the smartwatch, navigate between watch faces, and activate functions such as alarms or stopwatch. Button inputs are handled via GPIO pins on the ESP32.



Fig. 9. Tactile Push Buttons

IV. SOFTWARE DETAILS

The smartwatch system relies on software for sensor interfacing, data processing, display management, and IoT communication. The software framework is implemented using the Arduino IDE, which provides an extensive ecosystem of libraries for rapid development and hardware integration. Key software components include sensor data acquisition, signal processing, user interface management, IoT connectivity, and power management.

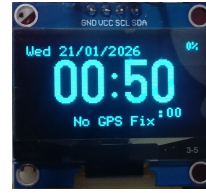


Fig. 10. Primary digital watch face showing real-time date, synchronized time, and GPS status

Time and System Synchronization: The system maintains accurate time using the RTC module, with optional synchronization through GPS or mobile connectivity. The primary digital interface displays real-time date and time along with GPS connection status, while an alternative analog interface visually represents time using clock hands for enhanced user familiarity.

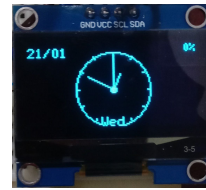


Fig. 11. Analog watch face representing real-time clock using graphical hour, minute, and second hands

Health Data Processing and Visualization: Physiological data is acquired from the MAX30102 sensor and processed to compute heart rate and blood oxygen saturation (SpO_2). Signal filtering and peak detection algorithms are applied to photoplethysmography (PPG) data to ensure reliable measurements. The processed health parameters are displayed in real time through a dedicated health interface.



Fig. 12. Health monitoring interface displaying real-time SpO₂ and heart rate values

Location and Environmental Information: The GPS module continuously provides geographic positioning data, including satellite count and fix status. This information is presented through a dedicated location interface, which also supports environmental or weather-related information when available. This feature enhances user safety and enables location-based monitoring through connected devices.



Fig. 13. Location interface showing GPS fix status and satellite information

Utility Functions- Alarm and Stopwatch: The software includes essential utility features such as an alarm and a stopwatch. The alarm module allows users to configure scheduled alerts using the button-based navigation system. The stopwatch function provides millisecond-level accuracy with start, stop, and reset operations, useful for daily activity tracking and timing tasks.



Fig. 14. Alarm configuration sub-menu for setting scheduled alerts



Fig. 15. Stopwatch interface with millisecond-accurate timing controls

System Configuration and Expansion Interface: A settings interface allows manual configuration of system parameters such as time, date, and operational modes. Additionally,

a placeholder game interface is included to demonstrate system expandability and future software enhancements without affecting core functionality.



Fig. 16. Settings interface for system configuration and manual adjustments

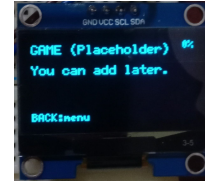


Fig. 17. Game placeholder interface for future entertainment module expansion

Power Management and Energy Optimization: To ensure extended battery life, the software implements power-saving strategies such as adaptive sensor polling and ESP32 deep-sleep modes. When the device enters low-power operation, a dedicated interface indicates the current sleep status of the system.



Fig. 18. Power-saving mode display indicating ESP32 deep-sleep status

V. PROTOTYPE

The smartwatch prototype integrates the ESP32 microcontroller, MAX30102 pulse oximeter, MPU6050 accelerometer, GPS and RTC modules, OLED display, and tactile buttons into a compact wearable device. It enables real-time monitoring of heart rate, blood oxygen saturation (SpO_2), and step/activity tracking, with data displayed on the OLED screen and transmitted via Wi-Fi for remote access.

The device features multiple watch faces displaying time, health metrics, location, and utility functions such as alarms and stopwatch. Designed for low power consumption and modularity, the prototype demonstrates the feasibility of a multifunctional IoT-enabled health monitoring smartwatch suitable for elderly and disabled users.

VI. CONCLUSION

In this project, a multifunctional IoT-enabled smartwatch has been designed and implemented to monitor vital signs and

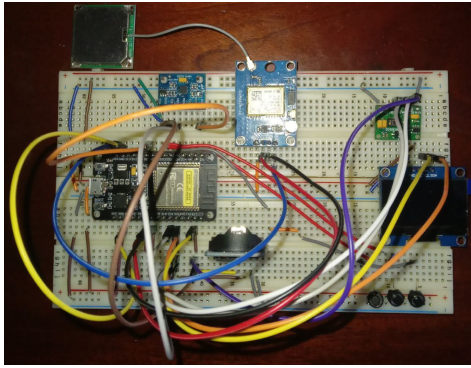


Fig. 19. Prototype of the Smartwatch



Fig. 20. Final assembled prototype of the Smartwatch

activity levels in real time. The system integrates the ESP32 microcontroller with key sensors such as the MAX30102 pulse oximeter, MPU6050 accelerometer, GPS, and RTC modules, providing a comprehensive health and lifestyle monitoring solution. The OLED display offers an intuitive interface for real-time feedback, while Wi-Fi connectivity allows data to be remotely accessed by caregivers or medical professionals.

The prototype demonstrates that a compact, wearable, and cost-effective device can provide continuous health monitoring for elderly and disabled individuals. The system is modular, energy-efficient, and scalable, with potential for future enhancements, such as additional physiological sensors or advanced analytics. Overall, the smartwatch combines real-time monitoring, IoT connectivity, and user-friendly operation to offer a practical tool for personalized healthcare.

VII. INDIVIDUAL CONTRIBUTIONS

- **Tasrina Hussan(2108014):** Designed and implemented the hardware prototype, including sensor integration and circuit assembly. Programmed the ESP32 for data acquisition and OLED display control.
- **Nadia Zaman(2108027):** Supervised the project, guided the system architecture, and provided technical support for sensor selection, data analysis, and documentation. Coordinated testing and evaluation of system performance.

- **Jannatul Maowa(2108049):** Developed the software framework for IoT connectivity, remote data visualization, and mobile application integration. Assisted with algorithm design for heart rate, SpO₂, and step count calculations.
- **Nafisa Nawar(2108058):** Assisted in prototype testing, debugging, and validation of sensor outputs. Contributed to the preparation of figures, documentation, and final report compilation.

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